CSIT212
QUANTITATIVE METHODS

COURSE ORIENTATION

First Semester 2627

Mattheus Marcus Contreras

CSIT212 - QUANTITATIVE METHODS COURSE ORIENTATION

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BS MATHEMATICS

DIPLOMA IN PROFESSIONAL EDUCATION

POST-BACCALAUREATE DIPLOMA IN COMPUTING

FREELANCE UI/UX & DATA ANALYST

CURRENTLY TAKING PROF MASTER IN DATA SCIENCE

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These help the institution appreciate its reason for being at the same time set directions for current undertakings and future endeavors.

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The **IDENTITY** is a general statement describing the nature of the institution and the core competency.

Cebu Institute of Technology – University is a **private, non-sectarian academic institution** founded in 1946. It provides **basic (early childhood, elementary, and high school), vocational, higher, and non-conventional or alternative education.**

Science & Technology is a way of academic life among administrators, faculty, and students. In the design, administration, and implementation of curricular programs, technology is embedded, articulated, and actuated.

Vision-Mission: TOPS AGAIN

Our **Vision** is to be a **Transdisciplinary** hub of excellence, learning, and innovation; **Outcomes-oriented** in philosophy; **Purpose-driven** in service to people and society; and a **Sustainability** champion **(T.O.P.S.)**.

In pursuit of this Vision, we commit to our **Mission** to **Advance excellence and innovation**; **Ground education in values and social responsibility**; **Anchor learning on outcomes**; **Integrate technology and sustainability**; and **Nurture people and partnerships (A.G.A.I.N.)**.

Core Principles: GEARUP

Growth, fostered by shared vision and innovation

Excellence, achieved through outcomes-based, technology-enhanced learning

Access, ensured via a safe, healthy, and supportive environment

Resilience, characterized by a regenerative and adaptive mindset

Unity, manifested through values-driven leadership

Professionalism, embodied in individual integrity and accountability

Core Values: CITU

Compassion & Care, we uphold empathy, inclusivity, and concern for all

Integrity & Innovation, we act with honesty and accountability, embracing creativity and technology

Teamwork & Tenacity, we value collaboration and resilience in facing challenges

Unceasing Quest for Excellence, we commit to lifelong learning, continuous improvement, and world-class standards

Graduate Attributes: TEKNOY

Transdisciplinary, applying critical thinking across fields
Ethical, upholding integrity and responsibility
Knowledgeable & Articulate, demonstrating expertise and clear communication
Nationalistic with a Global Mindset, balancing cultural rootedness with global awareness
Outcomes-oriented, delivering sustainable, agile results
Youthful, leading with energy and adaptability

COLLEGE OF COMPUTER STUDIES
VISION-MISSION-GOALS

VISION

A leading Center of Excellence in Computing Education

MISSION

To provide quality industry-standard computing education

GOALS

To produce values-driven, industry-preferred and competent computing professionals through: proficient, supportive and highly-trained faculty; appropriate facilities; programs and pedagogy responsive to the community; as well as a culture that encourages Creativity, Innovation, and Team-work

CSIT212 - QUANTITATIVE METHODS COURSE ORIENTATION

COURSE DESCRIPTION AND DETAILS

COURSE NAME | Quantitative Methods

COURSE CODE | CSIT212

NUMBER OF HOURS PER SEM | 54 HR

COURSE DESCRIPTION

This course emphasizes the use of statistical and mathematical analysis of data collected through questionnaires or surveys using various computational techniques. The data collected shall be processed using various mathematical models to provide a meaningful information which will be used by the management to make decisions to solve the occurrence of a problem.

COURSE DESCRIPTION AND DETAILS

REQUIREMENTS

- PC/Laptop/Mobile Device
- MS Office (Word, Excel)
- Yellow Paper (1 Whole) – For Activities
- Newsprint (Long) – For Quizzes
- Scientific Calculator
- Ruler (Optional)
- Pencil
- Ballpen (Black, Blue, Red)

COURSE OUTLINE

I. Introduction (3 hr)

- CIT-U and CCS VMGO
- Course Syllabus and Grading System

II. Introduction to Quantitative Analysis(6 hr)

- Overview of Quantitative Analysis
- The Quantitative Analysis Approach
- Developing a Quantitative Analysis Model
- Issues and Limitations in Quantitative Analysis

COURSE OUTLINE

III. Probability Concepts and Applications (9 hr)

- Fundamental Concepts of Probability
- Mutually Exclusive and Collectively Exhaustive Events
- Statistically Independent and Dependent Events

COURSE OUTLINE

IV. Decision Analysis (12 hr)

- Steps in the Decision-Making Process
- Types of Decision-Making Environments
- Decision Making under Uncertainty
- Decision Making under Risk
- Decision Trees and Their Applications

COURSE OUTLINE

V. Forecasting (12 hr)

- Steps in the Forecasting Process
- Types of Forecasting Techniques
- Measures of Forecast Accuracy
- Decomposition of Time Series Data
- Types of Time-Series Models
- Monitoring and Controlling Forecasts

COURSE OUTLINE

VI. Regression Analysis (12 hr)

- Scatter Diagrams and Data Visualization
- Simple Linear Regression
- Assessing the Fit of a Regression Model
- Significance Testing of Regression Models
- Multiple Regression Analysis
- Hypothesis Testing of Regression Coefficients

GRADING SYSTEM

Grading System Computation

1st Half (Midterm) = 10% (Prelim Exam) + 30% (Midterm Exam) + 60% (Class Standing)

2nd Half (Final) = 10% (Prelim Exam) + 30% (Midterm Exam) + 60% (Class Standing)

FG (Final Grade) = Average of 1st Half and 2nd Half

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GRADING SYSTEM

Course Requirements

- Class Attendance - Every Meeting (5 pts each)
- Activities (Seatworks / Homeworks / Recitations) - Every Meeting
- Quizzes (8 in total)
- Performance Tasks (1 per Grading Period)
- Major Examinations (Prelim, Midterm, Prefinal & Final)

Notes

Class Standing(60%) = Attendance(5%) + Activities(20%) + Quizzes(25%) + Perf Tasks(10%)

Transmutation Formula = ROUND(IF(GRADE% <= 60, 1 + (1 / 30) * GRADE%, 3 + (1 / 20) * (GRADE% - 60)),1)

GENERAL CLASSROOM & SCHOOL POLICIES

Regular Attendance

Attend all classes on time. Absences or tardiness require proper documentation and are subject to review.

Adhere to Rules

Follow the class schedule, procedures, and policies, including wearing the school uniform and ID.

GENERAL CLASSROOM & SCHOOL POLICIES

Engage in Coursework

Complete all activities, quizzes, tasks, and assignments.

Seek Clarification

Ask questions if anything is unclear. Address concerns through the proper channels.

GENERAL CLASSROOM & SCHOOL POLICIES

Study Habits

Develop effective study habits and manage your time wisely.

Take Care of Yourself

Maintain both physical and mental well-being.

“Can We Trust the Numbers?”

Question 1: What do you think is the average number of hours this class spends on their phones every day?

Question 2: Do students who spend more time on their phones have lower grades?

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Assignment

In a 1/2 cw, answer the following questions briefly. No need to copy the questions.

(1) Choose one concept in the Vision (TOPS) that you think is important to you and explain why.

(2) Choose one concept in the Mission (AGAIN) that you want to improve on and explain why.

(3) In the Core Principles (GEARUP), choose one that resonates with you and explain why.

(4) Which CITU Core Value would be the most significant for you? Why?

(5) What Graduate Attribute(TEKNOY) do you want to improve on? Why?

END OF ORIENTATION